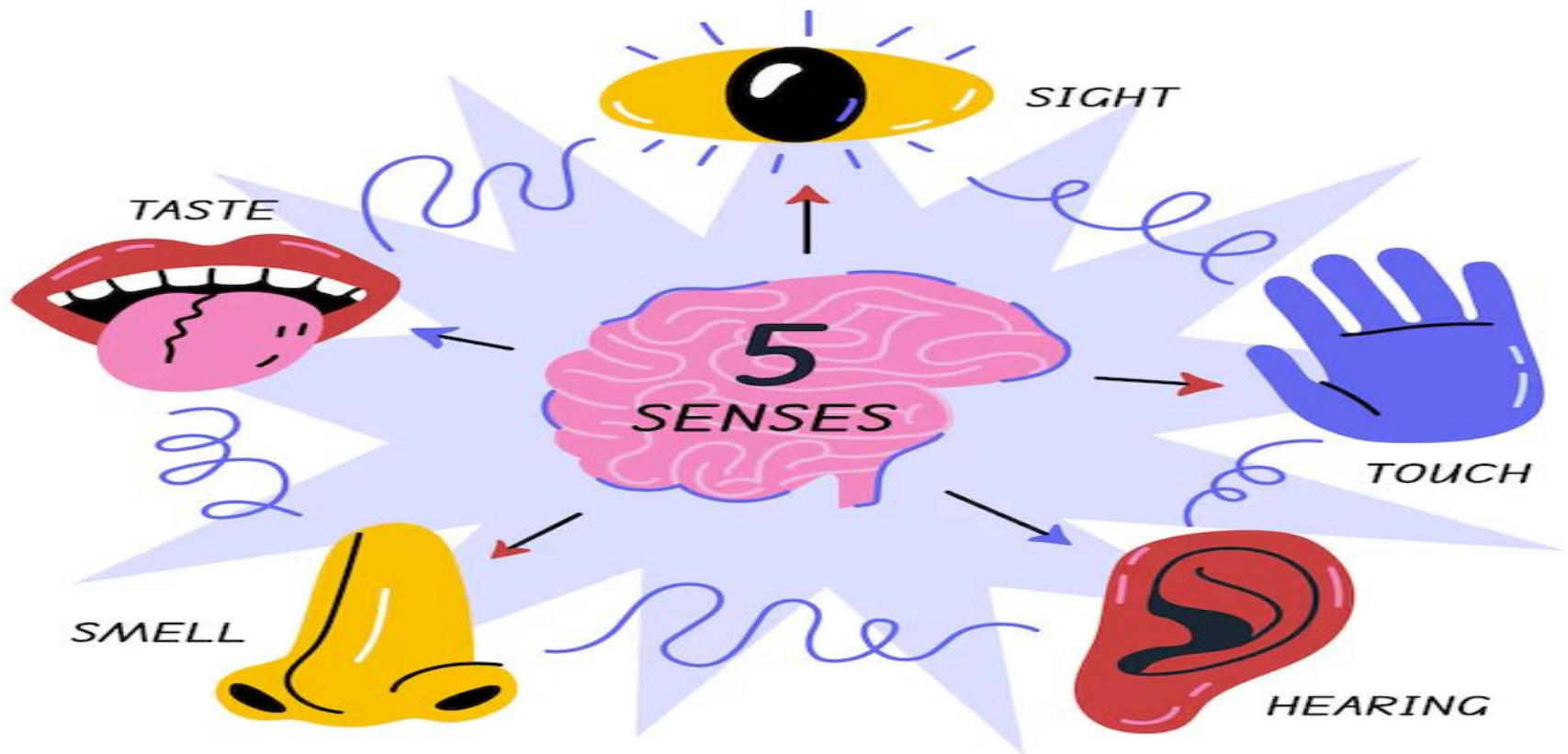


To Examine the Sensory systems of a human subject



DR ABDUL SAMAD KHAN
SENIOR LECTURER-DIKIOS
DOW UNIVERSITY OF HEALTH SCIENCES

The Neurological Examination





Neuro Exam Tools



128-H
Z



7 categories of the neurological exam

- Mental status
- Cranial nerves
- Motor system
- Reflexes
- Sensory system
- Coordination
- Station and gait

Station is the patient's attitude, posture, or manner of standing. The healthy individual stands erect with her/his head up, chest out, and abdomen in.

Sensory Examination

sensory pathway

Peripheral receptor



peripheral nerves



nerve roots



spinal cord



thalamus



sensory cortex

- Localisation of problems can be determined by knowledge of area of skin supplied by peripheral nerves, sensory dermatomes, decussation of spinothalamic tract and dorsal columns

DESCRIPTION:

Sensations are of two types; primary and cortical.

A-Primary sensations Somatic Sensation

It includes touch, pain, temperature, position, passive movement and vibration.

B-Cortical sensations are localization of two point discrimination, stereognosis and graphesthesia

Examination Sequence

- Touch
- Pain
- Deep pain
- Temperature
- Joint position sense
- Vibration sense
- Two-point discrimination

Sensory Exam: General Points

- Explain each test before you do it
- Patient's eyes should be closed during testing
- Test all 4 extremities
- Avoid leading questions like “Is this sharp?”
- Compare side-to-side and distal-to-proximal asking if they are “about the same”
- When you detect an area of sensory loss, map out its boundaries in detail

Components of Sensory examination

1-Primary sensation OR Somatic Sensation

Somatic sensation is the process where your body detects physical experiences like touch, temperature, pain, and body position.

2-A cortical sensation is a higher-level sensory function that requires the cerebral cortex (specifically the parietal lobe) to process, interpret, and discriminate complex sensory stimuli. It depends on basic primary sensations like Two point discrimination, Stereognosis and graphesthesia.

3-Hearing test

Webber's Test, Rinse Test

4-Vision

Pupillary light reflex, Ishihara chart

Characterize as normal, absent, reduced, exaggerated, or perverted
(*dysesthesias*)

- **Integrative sensation**

- Graphesthesia
- Stereognosis
- Double simultaneous stimulation

Primary OR Somatic Sensation

1- Sense of Touch

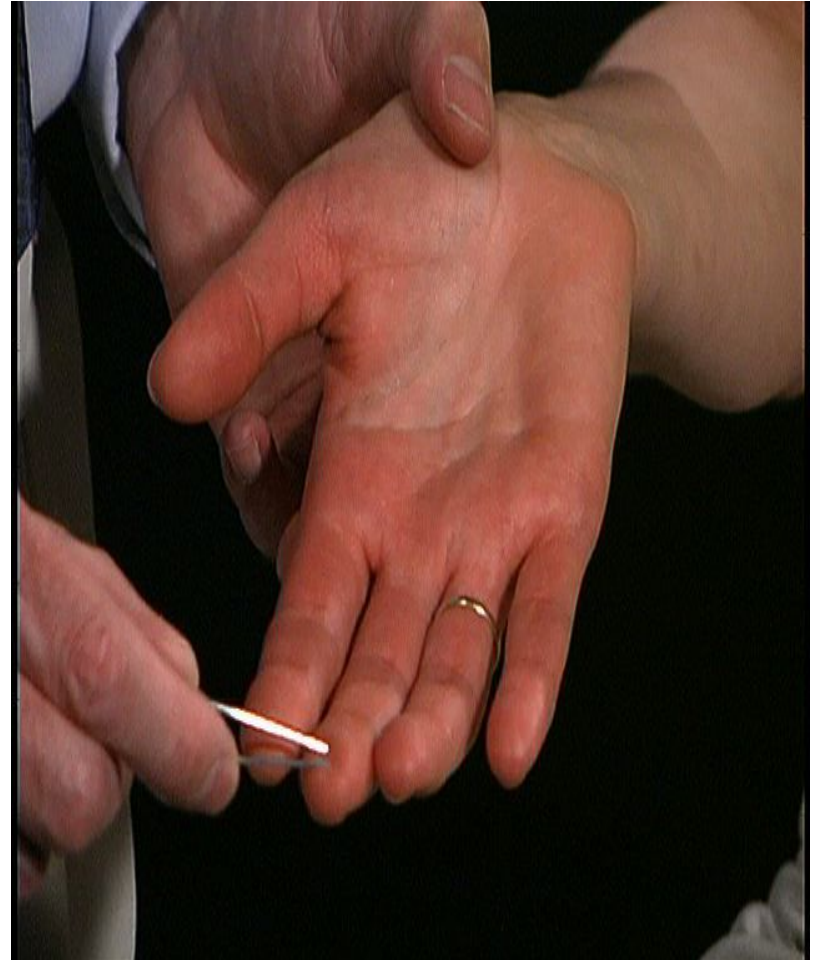
Touch the skin with cotton wool or a piece of paper. Ask the pt to close the eyes and raise the finger, say yes or no when the pt feels or does not feel the touching object.

Following are the abnormalities of the sense of touch:

Anesthesia: Touch is completely lost

Hypoesthesia: Decreases sensation of touch

Hyperesthesia: Sensation of touch is aggravated.



Light Touch & Double Simultaneous Stimulation

- Lightly touch face and extremities in random order, asking patient to respond whenever a touch is felt
- Touch both sides of face or body simultaneously, asking patient to indicate whether touch is felt on left, right, or both sides



2- Sense of Pain

Ask the pt to close the eyes, touch with the sharp blunt ends of the pain in random sequence and see whether the pt can differentiate between the two or not. If the subject feel but can not distinguish between sharp and blunt, it means sensation of touch is intact, but sensation of pain is lost.



Pain sensation

1-Test for distal gradient of sensory loss in the leg

2-Test for sensory loss in most commonly affected nerve and nerve root distributions

- Palmar aspect of index finger (median nerve).
- Palmar aspect of 5th finger (ulnar nerve)
- Web space between thumb and index finger on dorsal surface of hand (radial nerve).
- Lateral surface of foot (L5).
- Posterior aspect of leg (S1).

Ask patient if the sensation is “about the same”



Sense of Temperature:

Take two test tubes containing cold and warm water and apply to skin and ask the subject to indicate whether it is hot or cold



4-Sense of position

Ask the person to close his eyes. Now move his limb in various directions and then place it in some position. Ask the person to place the other limb in the same position. If sense of position is intact, he will bring the other limb in exactly the same position.



5-Sense of vibration:

To test this sense ,place the vibrating tuning fork over some bony prominence,e.g tibial tuberosity or medial or lateral malleoli. Commonly the person feels the buzzing sensation.



B-Cortical Sensation

Somatosensory area is responsible for exact localization and discrimination of primary sensations and damage to this area results in loss of two points discrimination, stereognosis and paragnathesis.

Make sure pt is normal and speech and primary sensations are intact.

1- Sense of localization :- Touch a part of pt body e.g. face with eyes closed. Then ask him to localize the part of his body and then ask the pt to open the eyes and place his finger on that part.

2-Two point discrimination.

The two-point discrimination test measures your ability to feel two close points on your skin as separate touches instead of just one. Doctors use it to check nerve damage, track nerve recovery, or evaluate brain pathways that handle touch.

How to measure 2-point discrimination?

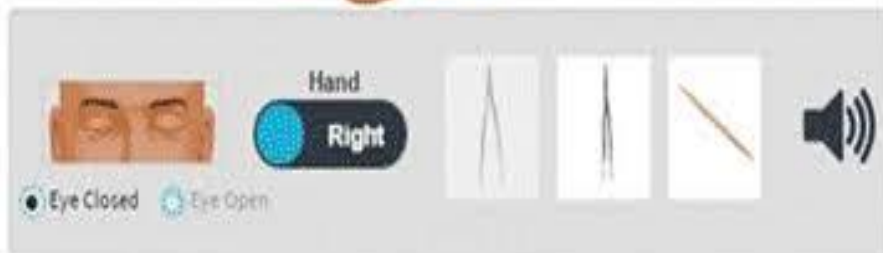
To measure two-point discrimination, use a specialized tool like a two-point discriminator, a caliper, or a bent paperclip to touch the skin with one or two points, asking the person to identify if they feel "one" or "two" points.

A normal result on a fingertip is 5 mm or less.

How the Test Works?

- The person being tested closes their eyes.
- A tool with two blunt points (like a caliper or a special disk) touches the skin.
- The tester switches between using one point or two points.
- You must say if you feel one touch or two.
- The tester moves the points closer or farther apart to find the smallest gap you can tell
- Normal Values by Body Area
- Different parts of the body have different numbers of touch sensors.
- Sensitive areas need a very small gap, while less sensitive areas need a large gap.
- Fingertips: Less than 6 mm (often 2 to 5 mm)
- Palms: 8 to 12 mm
- Back or Shins: 30 to 40 mm

Two Point Discrimination



Graphesthesia is the ability of the patient to identify characters that are written on the skin using a dull pointed object.



Graphesthesia

Stereognosis is the ability to identify objects that are placed in the hand when the eyes are closed.



Stereognosis

Graphesthesia



recognition of number
or letter traced on the
skin

Stereognosis



ability to recognize the
form of objects by touch

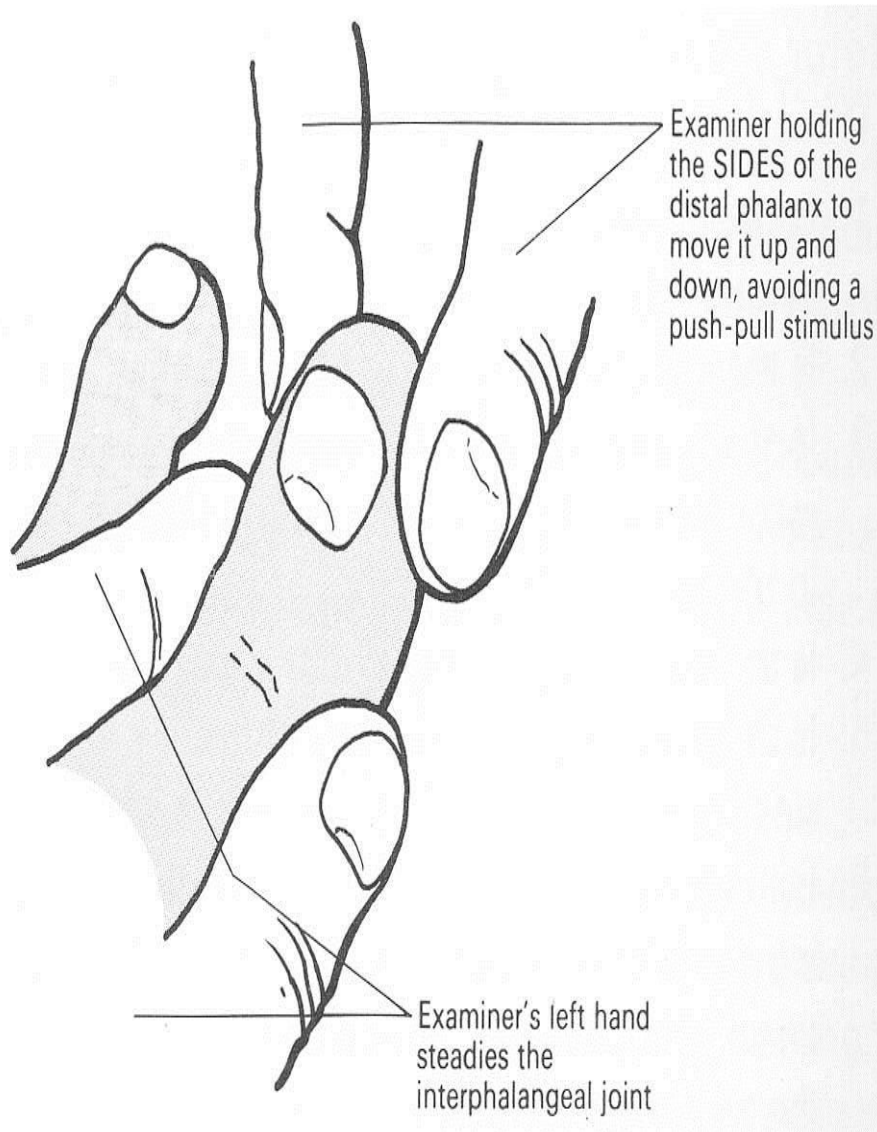
Baragnosis



ability to recognize
weight of an object



Joint position sense



- Test toes and fingers
- Move digit only a few degrees
- If impaired, move digit greater distance -> test more proximally

Sensory deficit lesions

- Sensory deficit lesions are areas of damage in the nervous system that disrupt how the body perceives touch, pain, temperature, vibration, or proprioception
- Lesions causing sensory deficits involve the nervous system or skin pathways.
- Location dictates the symptom pattern:
 - peripheral nerve damage causes localized numbness,
 - spinal cord lesions create dissociated or level-specific losses, and
 - brain lesions trigger contralateral hemisensory deficits

Peripheral Nerve and Skin Lesions

Single nerve (Mononeuropathy). (e.g., Carpal Tunnel Syndrome):

Carpal tunnel syndrome is a common medical condition caused by pressure on the median nerve as it passes through a narrow, rigid passageway of ligament and bones in the wrist. This compression leads to numbness, tingling, burning, or weakness in the hand and

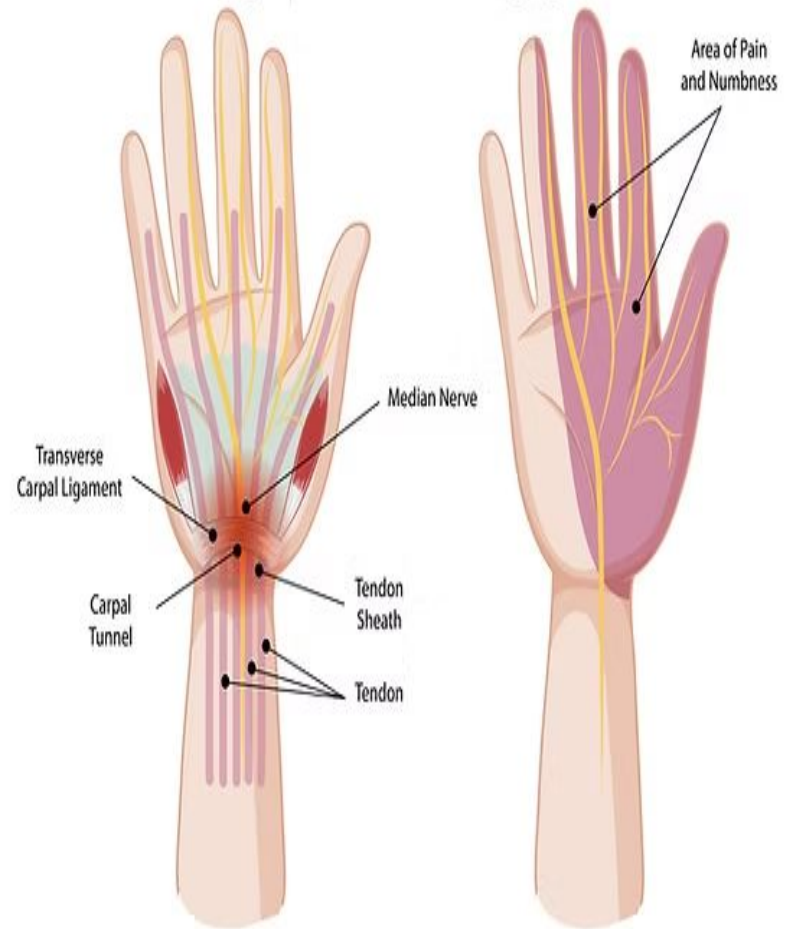
Small-fiber polyneuropathy: Selective loss of small nerve fibers leading to burning pain and impaired thermal sensation with normal vibration/touch

Nerve root (Radiculopathy): Loss occurs in a specific skin strip or dermatome, often caused by a slipped spinal disc pressing on a nerve.

Multiple nerves (Polyneuropathy): "Stocking-glove" numbness starts first in the longest toes and fingers, common in diabetes

Carpal Tunnel Syndrome

Compression of the median nerve as it passes through the wrist at the carpal tunnel, leading to pain, numbness, and tingling in the hand.

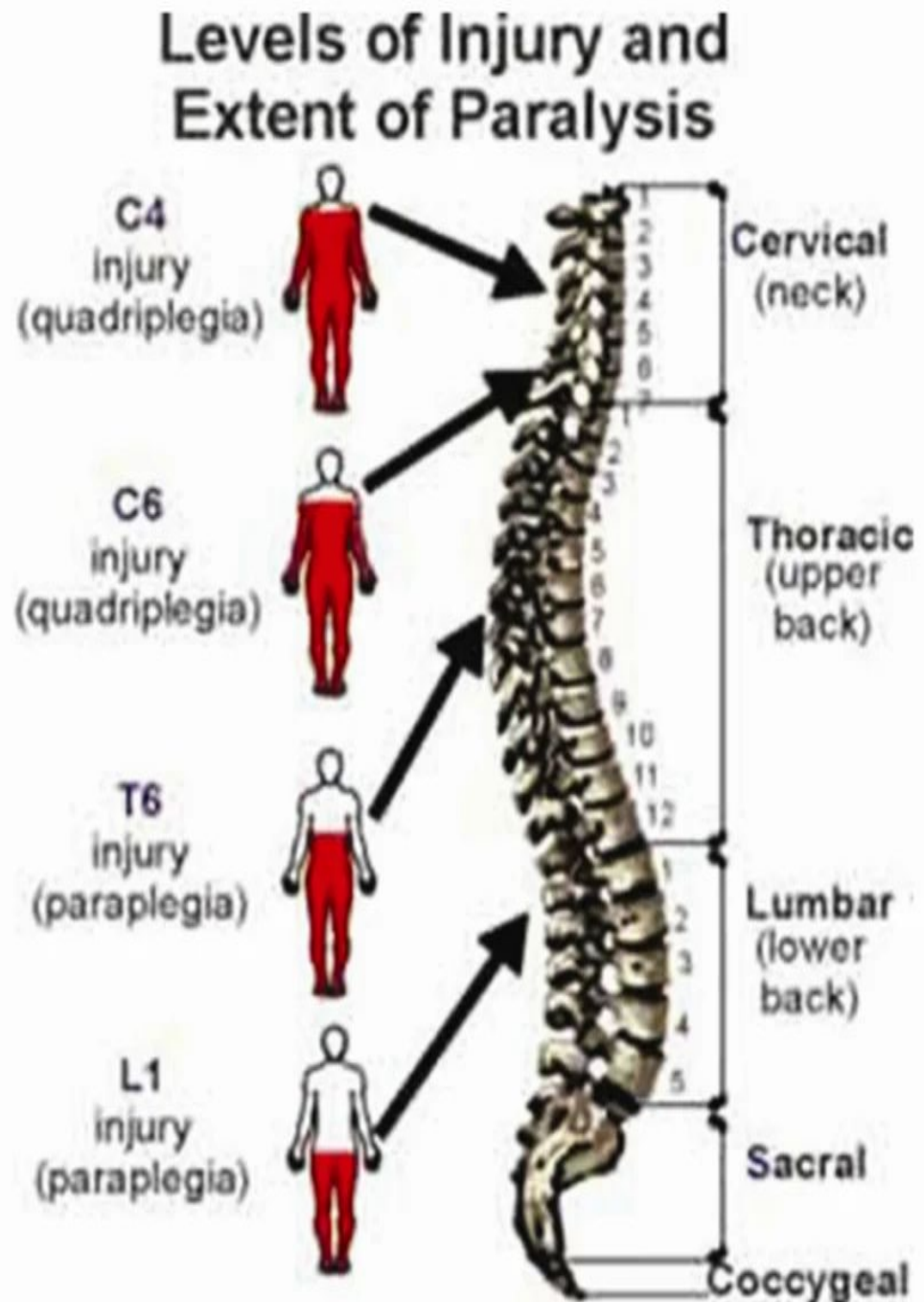


Spinal Cord Lesions

Complete transection: Total loss of all feeling and movement below the level of the damage.

Brown-Séquard syndrome (Half-cord damage): Loss of pain and temperature on the opposite side, but loss of vibration and position sense on the same side as the injury.

Central cord syndrome (Syringomyelia/Tumor): "Dissociated sensory loss" where pain and temperature are lost across the shoulders and arms, but touch and vibration stay normal.



BACKGROUND

- * AREA of SKIN INNERVATED by SPECIFIC NERVE ROOT
- ~ 30 TOTAL

8 CERVICAL

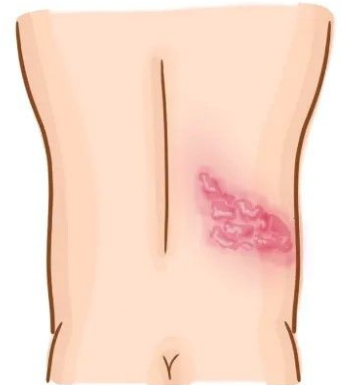
12 THORACIC

5 LUMBAR

5 SACRAL

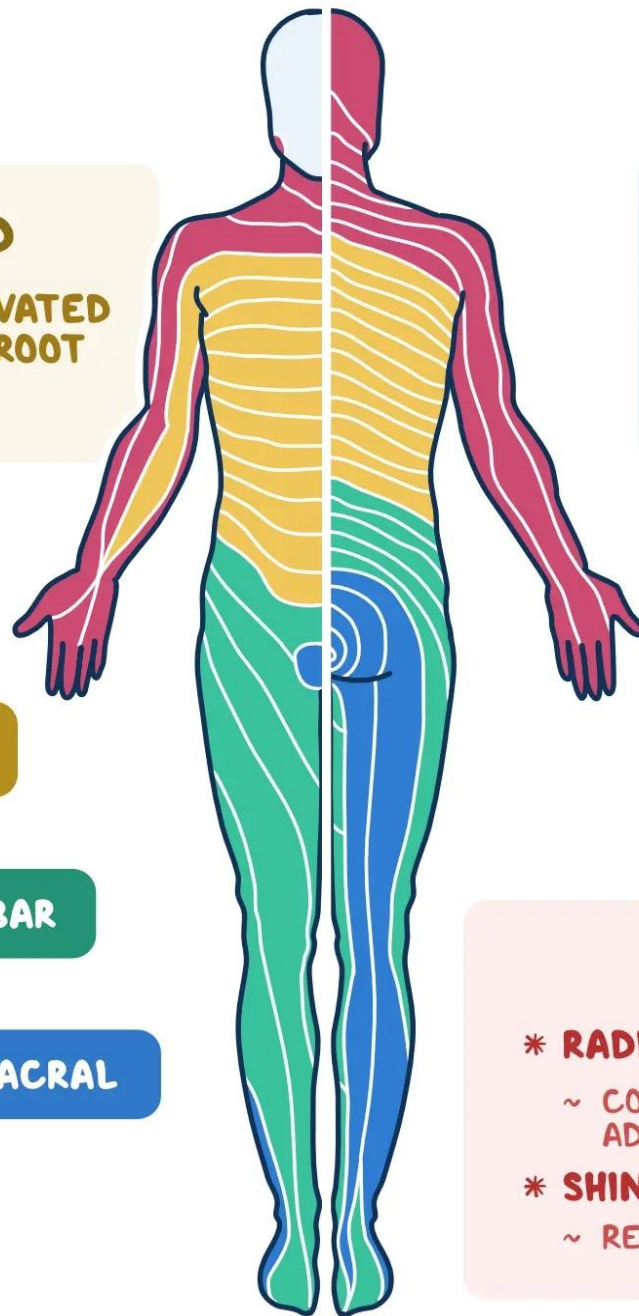
ASSESSMENT

- * NEUROLOGICAL EXAM
 - ~ PINPRICK TEST
 - ~ LIGHT TOUCH TEST



PATHOLOGIES

- * **RADICULOPATHIES**
 - ~ COMMONLY due to DISC HERNIATIONS from ADVANCED AGE or TRAUMA
- * **SHINGLES**
 - ~ REACTIVATION of VARICELLA ZOSTER VIRUS



ANTERIOR

POSTERIOR

Brainstem and Thalamus Lesions

Lateral Medullary (Wallenberg) Syndrome:

Brainstem stroke causing ipsilateral facial pain/temperature loss and contralateral body pain/temperature loss. Crossed findings, such as numbness on one side of the face and the opposite side of the body.

Brainstem stroke: Thalamic stroke:

Severe or burning pain and altered sensation over an entire half of the body (contralateral hemisensory loss).

Cerebral Cortex Lesions

Parietal lobe damage:

The patient loses fine sensory discrimination (like recognizing shapes or objects by touch) on the opposite side of the body, while basic pain and temperature detection remain intact.



PARIETAL LOBES

The parietal lobe rests near the top and center of the cerebral cortex, just behind the frontal lobe and above the occipital and temporal lobes.

IT IS VITAL FOR SENSORY PERCEPTION AND INTEGRATION



SIGHT



TASTE



HEARING



SMELL



TOUCH

SOME IMPORTANT FUNCTIONS OF THE PARIETAL LOBE



COGNITION



SPEECH



WRITING



Relevant Questions

Q-1-How would you characterize the touch receptor distribution pattern in the skin location you tested

The distribution pattern of touch receptors is characterized by high variability across different skin locations, with a higher density of receptors in areas requiring greater tactile acuity, such as the fingertips and palms, and lower density in hairy skin on the arms, legs, and face. This differential distribution allows for fine discrimination of light touch, pressure, and vibration in sensitive areas and a broader detection of stimuli in less sensitive regions.

Q-2-How does the distribution of touch receptors appear to differ between regions?

Touch receptor distribution varies significantly across the body; highly sensitive areas like the fingertips, lips, and tongue have very high densities of receptors, resulting in high tactile resolution and small receptive fields. In contrast, less sensitive regions, such as the back and arms, have lower densities of touch receptors, larger receptive fields, and lower tactile resolution, allowing them to sense touch over broader areas but with less precision. This uneven distribution is often correlated with the presence or absence of hair and the functional requirements of each body part.

Q-3-What area of the skin has the greatest and least ability to differentiate between two points?

The skin areas with the greatest ability to differentiate two points are the fingertips, lips, and tongue, while the back has the least ability. This difference is due to the density of touch receptors, such as Merkel's disks and Meissner's corpuscles, with highly sensitive areas having more receptors and smaller receptive fields, allowing for more precise tactile perception.

Q-4-What do you understand by somatic senses?

Somatic senses, also known as the somatosensory system, are the general sensations that originate from the body, including touch, pressure, vibration, temperature, and pain. These sensations are detected by specialized receptors in the skin and throughout the body and are communicated to the brain via sensory neurons, allowing for perception of the external environment and the body's position in space

Q-5-What is the importance of examining sensory system?

Examining the sensory system is important for diagnosing and locating neurological problems by assessing a patient's ability to sense touch, pain, temperature, and position. This helps determine the extent and severity of sensory impairments and informs subsequent diagnostic steps like imaging or biopsies. In clinical medicine, understanding the sensory system is vital for treating conditions like chronic pain, and in areas like pediatrics, it's crucial for assessing learning and development.

Q-6-Differentiate between primary and cortical sensation-

Primary sensations are basic sensory inputs like touch, pain, temperature, and proprioception, which are transmitted directly from the body to the brain's primary sensory cortex for initial processing. In contrast, cortical sensations are higher-order cognitive abilities that involve more complex processing within the cortex, requiring the brain to interpret and integrate these primary inputs to perform tasks like recognizing an object by touch (stereognosis), identifying letters written on the skin (graphesthesia), or discriminating between two nearby points of touch (two-point discrimination).

Q-7-What is Astereognosis and graphesthesia?

Stereognosis is the ability to recognize an object by touch alone, while graphesthesia is the ability to recognize symbols or numbers written on the skin by touch. Both are forms of tactile sensation and rely on the integrity of the somatosensory cortex. Astereognosis (loss of stereognosis) or graphanesthesia (loss of graphesthesia) can indicate neurological dysfunction, often in the parietal lobe.

Q-8-In hemisection of spinal cord, which sensations are lost on which sides of body?

In hemisection of the spinal cord, known as Brown-Séquard syndrome, there is ipsilateral (same side) loss of motor function, proprioception, vibration, and fine touch, and contralateral (opposite side) loss of pain and temperature sensation below the level of the lesion. For example, a right-sided hemisection results in paralysis and loss of position sense on the right side, while pain and temperature are lost on the left side.